BRSM Movie Memory Experiment

Comprehensive Statistical Report

After Vigilance-Based and Repeated-Video Exclusions

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Contents

1 Overview and Data Quality

1.1 Vigilance-based exclusions

Two participants failed the embedded vigilance check and are excluded from every analysis:

- `sub105_AB`
- `sub70_NB`

1.2 Repeated-video exclusion

Five movies per condition (IDs 3, 7, 18, 28, 37) were presented *twice* during encoding as attention checks. All recognition trials corresponding to these repeated movies are excluded from every analysis. **Rationale:** participants received two encoding exposures for these clips; their recognition performance is therefore not comparable to single-exposure movies and would artificially inflate accuracy estimates.

1.3 Sample and trial structure

The retained analytical sample is $N = 168$ (NB=88, AB=80), contributing 5,880 recognition trials (35 unique movies $\times$ 168 participants).

Condition	n	n trials	BB trials	EM trials
NB (Natural Boundary)	88	3,080	1540	1540
AB (Abrupt Boundary)	80	2,800	1400	1400
Total	168	5,880	—	—

Table 1: Analytical sample after all exclusions (35 unique movies per participant).

2 Statistical Methods

Each hypothesis follows an identical pre-specified decision pipeline:

1. Build the DV at the participant level from trial-level data.
2. Compute full descriptive statistics (n, M, SD, Mdn, Q_1, Q_3, min, max) per group.
3. Test normality within each group using the Shapiro–Wilk test ($\alpha = .05$).
4. Test variance homogeneity using Levene's test (median-centred; Brown–Forsythe variant).
5. Select the inferential test:
   - Both groups normal *and* equal variance $\Rightarrow$ Student's t-test
   - Both groups normal *and* unequal variance $\Rightarrow$ Welch's t-test
   - Either group non-normal $\Rightarrow$ Mann–Whitney U test (two-sided)
6. Report effect sizes: Cohen's d / Hedges' g for parametric tests; rank-biserial r for non-parametric tests.

No correction for multiple comparisons is applied. All reported p-values are raw.

2.1 Accuracy dependent variable

All accuracy-based DVs use raw proportion correct (range 0–1). Chance performance is 0.5 in a two-alternative forced-choice task. After excluding 5 repeated movies per participant, each participant contributes 35 unique-movie trials (approximately 17–18 BB + 17–18 EM).

2.2 Log-transformation and normality of response time

Trial-level RTs are strongly right-skewed (skew ≈ 3.85, excess kurtosis ≈ 32.50). All RT analyses therefore use natural-log-transformed RT (ln seconds). Table 2 verifies that the transform rescues approximate normality at the participant level.

Condition	Transform	n	M	Shapiro–Wilk p
NB	Raw RT (s)	88	5.883 s	$< .001$ (non-normal)
	Log RT (ln s)	88	1.626	.5024 (normal)
AB	Raw RT (s)	80	5.708 s	$< .001$ (non-normal)
	Log RT (ln s)	80	1.595	.0746 (normal)
All (trial)	Raw RT	5000 sample	skew= 3.853	$< .001$
	Log RT	5000 sample	skew= 0.588	$< .001$

Table 2: Shapiro–Wilk normality: raw RT vs. log RT at participant level (repeated movies excluded).

Note. At the trial level ($n \approx 5,880$) Shapiro–Wilk has extreme power and flags even trivial departures from normality. The skew drops substantially after log-transformation, confirming normalisation. At the participant level both conditions pass after log-transformation.

3 H1: Overall Recognition Accuracy

3.1 Hypotheses

H_0: Overall recognition accuracy does not differ between Natural Boundary and Abrupt Boundary participants ($\mu_{\text{NB}} = \mu_{\text{AB}}$).

H_1: NB participants have higher overall recognition accuracy than AB participants ($\mu_{\text{NB}} > \mu_{\text{AB}}$; directional).

3.2 DV and descriptive statistics

DV: Per-participant proportion correct across all 35 unique recognition trials (repeated movies excluded).

Group	n	M	SD	Mdn	Q1	Q3	Min	Max
NB	88	0.867	0.075	0.886	0.829	0.921	0.629	1.000
AB	80	0.832	0.084	0.829	0.771	0.886	0.571	1.000

Table 3: Descriptive statistics — H1 (Overall accuracy).

3.3 Test selection and result

Normality (Shapiro–Wilk): NB $p = .0028$; AB $p = .0343$. At least one group rejects normality.

Variance homogeneity (Levene): $p = .3803$. Equal variances assumed.

Test selected: Mann–Whitney U. Tied ranks: 12 groups from 16 distinct values (tie correction = 0.9865, negligible).

Result: $U = 4384.500$, $p_{\text{one-sided}} = .0029$. Rank-biserial r = 0.246.

Decision: Reject $\mathbf{H_0}$. NB participants have significantly higher overall accuracy.

4 H2: Before-Boundary Frame Accuracy

4.1 Hypotheses

H_0: Recognition accuracy for Before-Boundary targets does not differ between conditions ($\mu_{\text{NB}} = \mu_{\text{AB}}$).

H_1: NB participants have higher Before-Boundary accuracy than AB participants ($\mu_{\text{NB}} > \mu_{\text{AB}}$; directional).

4.2 DV and descriptive statistics

DV: Per-participant proportion correct on BB-frame targets only (approximately 17–18 trials after exclusions).

Group	n	M	SD	Mdn	Q1	Q3	Min	Max
NB	88	0.854	0.100	0.889	0.778	0.944	0.556	1.000
AB	80	0.808	0.115	0.833	0.722	0.889	0.389	1.000

Table 4: Descriptive statistics — H2 (BB accuracy).

4.3 Test selection and result

Normality (Shapiro–Wilk): NB $p =< .001$; AB $p =< .001$. At least one group rejects normality.

Variance homogeneity (Levene): $p = .4261$. Equal variances assumed.

Test selected: Mann–Whitney U. Tied ranks: 9 groups from 11 distinct values (tie correction = 0.9729, negligible).

Result: $U = 4370.000$, $p_{\text{one-sided}} = .0031$. Rank-biserial r = 0.241.

Decision: Reject $\mathbf{H_0}$. NB participants recognise BB-frame targets significantly more accurately.

5 H3: Mean Confidence Rating

5.1 Hypotheses

H_0: Mean confidence rating does not differ between NB and AB participants ($\mu_{\mathrm{NB}} = \mu_{\mathrm{AB}}$).

H_1: NB participants report higher mean confidence than AB participants ($\mu_{\mathrm{NB}} > \mu_{\mathrm{AB}}$; directional).

5.2 DV and descriptive statistics

DV: Per-participant mean confidence rating (1–5 Likert scale; repeated movies excluded).

Group	n	M	SD	Mdn	Q1	Q3	Min	Max
NB	88	4.177	0.422	4.243	3.907	4.521	2.971	5.000
AB	80	4.040	0.476	4.043	3.650	4.407	2.829	4.886

Table 5: Descriptive statistics — H3 (Mean confidence).

5.3 Test selection and result

Normality (Shapiro–Wilk): NB $p = .0312$; AB $p = .1607$. At least one group rejects normality.

Variance homogeneity (Levene): $p = .1211$. Equal variances assumed.

Test selected: Mann–Whitney U. Tied ranks: 46 groups from 58 distinct values (tie correction = 0.9994, negligible).

Result: $U = 4097.500$, $p_{\text{one-sided}} = .0334$. Rank-biserial r = 0.164.

Decision: Reject $\mathbf{H_0}$. NB participants report significantly higher mean confidence.

6 H4: Proportion of Low-Confidence Trials

6.1 Hypotheses

H_0: The proportion of trials with confidence ≤ 3 does not differ between conditions ($\pi_{\mathrm{AB}} = \pi_{\mathrm{NB}}$).

H_1: AB participants have a higher proportion of low-confidence trials ($\pi_{\mathrm{AB}} > \pi_{\mathrm{NB}}$; directional).

6.2 DV and descriptive statistics

DV: Per-participant proportion of trials rated confidence ≤ 3 (repeated movies excluded).

Group	n	M	SD	Mdn	Q1	Q3	Min	Max
AB	80	0.277	0.153	0.286	0.164	0.400	0.000	0.657
NB	88	0.243	0.132	0.229	0.143	0.343	0.000	0.629

Table 6: Descriptive statistics — H4 (Low-confidence proportion).

6.3 Test selection and result

Normality (Shapiro–Wilk): AB $p = .1431$; NB $p = .0711$. Both groups pass normality.

Variance homogeneity (Levene): $p = .0922$. Equal variances assumed.

Test selected: Student's independent-samples t-test.

Result: $t = 1.577$, $df = 166$, $p_{\text{one-sided}} = .0584$. Cohen's d (pooled) = 0.244.

Decision: Fail to reject H_0. AB participants show numerically more low-confidence responses.

7 H5: Per-Movie Accuracy Variability

7.1 Hypotheses

H_0: The standard deviation of per-movie accuracy does not differ between conditions ($\sigma_{\text{AB}} = \sigma_{\text{NB}}$).

H_1: AB participants show greater variability in per-movie accuracy ($\sigma_{\text{AB}} > \sigma_{\text{NB}}$; directional).

7.2 DV and descriptive statistics

DV: Per-participant standard deviation of accuracy across the 35 unique movies (repeated movies excluded).

Group	n	M	SD	Mdn	Q1	Q3	Min	Max
AB	80	0.360	0.085	0.382	0.323	0.426	0.000	0.502
NB	88	0.324	0.089	0.323	0.272	0.382	0.000	0.490

Table 7: Descriptive statistics — H5 (Per-movie accuracy SD).

7.3 Test selection and result

Normality (Shapiro–Wilk): AB $p =< .001$; NB $p = .0017$. At least one group rejects normality.

Variance homogeneity (Levene): $p = .5266$. Equal variances assumed.

Test selected: Mann–Whitney U. Tied ranks: 12 groups from 16 distinct values (tie correction = 0.9865, negligible).

Result: $U = 4428.000$, $p_{\text{one-sided}} = .0019$. Rank-biserial r = 0.258.

Decision: Reject $\mathbf{H_0}$. AB participants show significantly greater per-movie accuracy variability.

8 H6: BB-Frame vs EM-Frame Response Time

8.1 Hypotheses

H_0: Mean log RT does not differ between BB-frame and EM-frame recognition trials ($\mu_{BB} = \mu_{EM}$).

H_1: Mean log RT differs between BB-frame and EM-frame recognition trials ($\mu_{BB} \neq \mu_{EM}$; non-directional).

Rationale. If BB-frame targets engage a qualitatively different retrieval process (e.g., recollection rather than familiarity), recognition decisions for BB targets may take longer than for EM targets. This within-participant comparison is exploratory and non-directional.

8.2 DV and descriptive statistics

DV: Per-participant mean log RT on BB trials and mean log RT on EM trials; the within-participant difference BB−EM is tested against zero (repeated movies excluded).

Measure	n	M	SD	Mdn	Q1	Q3	Min	Max
BB mean log RT (ln s)	168	1.635	0.243	1.615	1.491	1.756	1.045	2.340
EM mean log RT (ln s)	168	1.586	0.246	1.596	1.416	1.741	1.029	2.290
BB−EM difference	168	0.048	0.148	0.054	-0.045	0.160	-0.329	0.426

Table 8: Descriptive statistics — H6 (BB vs EM log RT, within-participant).

8.3 Test selection and result

Normality of differences (Shapiro–Wilk): $p = .6268$ (normal). One-sample t-test on differences is used.

Test selected: One-sample t-test on differences (paired, two-tailed).

Result: $t = 4.256$, $df = 167$, $p_{\text{two-sided}} =< .001$. Effect: Cohen's d_z = 0.328.

Decision: Reject H_0. BB-frame recognition decisions take significantly longer than EM-frame decisions (mean difference = 0.048 ln s), consistent with a more effortful retrieval process for boundary-region events.

9 H7: Mean log RT vs Overall Accuracy

9.1 Hypotheses

H_0: There is no linear association between mean response time and overall recognition accuracy ($\rho = 0$; two-tailed).

H_1: There is a significant association between mean response time and overall recognition accuracy ($\rho \neq 0$; two-tailed).

Rationale. Slower responses on a recognition task could reflect greater uncertainty and more effortful retrieval, which may be associated with lower accuracy (speed–accuracy trade-off). Alternatively, slower, more careful responding could improve accuracy. The direction is not pre-specified; the test is two-tailed.

9.2 DV and descriptive statistics

DV: Per-participant mean log RT (ln s) and per-participant overall accuracy (proportion correct across all 35 unique trials; repeated movies excluded). Log RT is used because raw RT is right-skewed (see §2.3).

Variable	n	M	SD	Mdn	Q1	Q3	Min	Max
Mean log RT (ln s)	168	1.611	—	—	—	—	—	—
Overall accuracy	168	0.851	—	—	—	—	—	—

Table 9: Descriptive statistics — H7 (Mean log RT and Overall accuracy).

9.3 Test selection and result

Normality (Shapiro–Wilk): Mean log RT: $W = 0.982$, $p = .0262$; Overall accuracy: $W = 0.960$, $p =< .001$.

Test selected: Spearman rank correlation (at least one variable non-normal; two-tailed). 95% CI obtained by bootstrap ($B = 5000$).

Result: $\rho = -0.036$ ($n = 168$, $df = 166$), $p_{\text{two-sided}} = .6429$, 95% CI $[-0.196, 0.116]$.

Decision: Fail to reject H_0. No significant correlation was found between mean log RT and overall accuracy ($\rho = -0.036$, $p = .6429$).

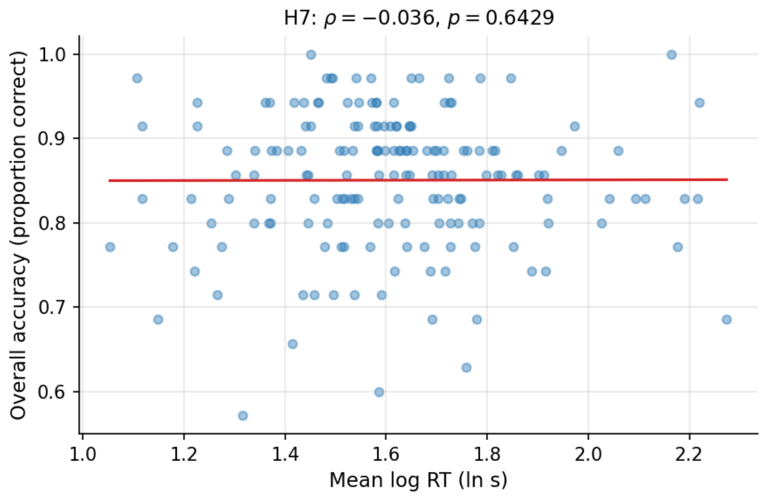

Figure 1: H7 scatter plot: per-participant mean log RT vs. overall recognition accuracy (repeated movies excluded). Regression line added.

10 H8: Mean log RT vs Mean Confidence Rating

10.1 Hypotheses

H_0: There is no linear association between mean response time and mean confidence rating ($\rho = 0$; two-tailed).

H_1: There is a significant association between mean response time and mean confidence rating ($\rho \neq 0$; two-tailed).

Rationale. High confidence is associated with fast, fluent recognition (the 'feeling of knowing' literature), so slower mean RT may predict lower confidence. The direction is not pre-specified; the test is two-tailed.

10.2 DV and descriptive statistics

DV: Per-participant mean log RT (ln s) and per-participant mean confidence rating (1–5 Likert scale). Log RT is used for the same reason as H7. Repeated movies are excluded.

10.3 Test selection and result

Normality (Shapiro–Wilk): Mean log RT: $p = .0262$; Mean confidence: $p = .0053$.

Test selected: Spearman rank correlation (at least one variable non-normal; two-tailed). 95% CI obtained by bootstrap ($B = 5000$).

Result: $\rho = -0.163$ ($n = 168$, $df = 166$), $p_{\text{two-sided}} = .0342$, 95% CI $[-0.313, -0.005]$.

Decision: Reject H_0. A statistically significant negative Spearman correlation was found between mean log RT and mean confidence rating ($\rho = -0.163$, $p = .0342$).

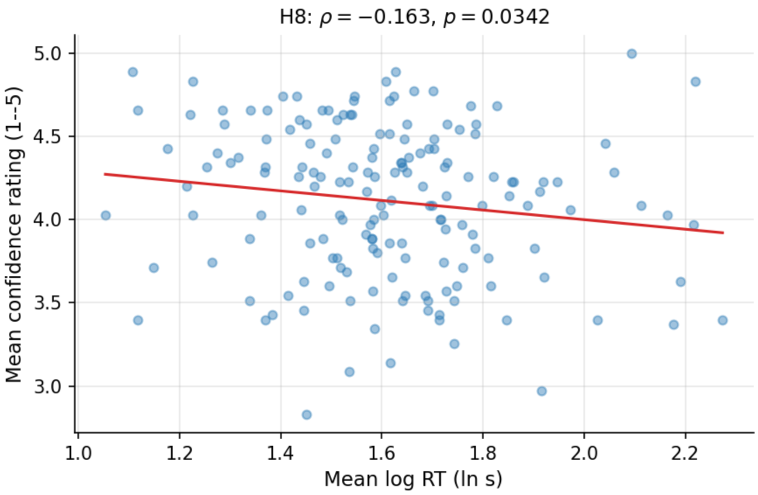

Figure 2: H8 scatter plot: per-participant mean log RT vs. mean confidence rating (repeated movies excluded). Regression line added.

11 Regression Analysis

11.1 Overview and Feature Policy

A strict regression pipeline was applied to predict trial-level recognition accuracy (`resp.corr`, binary) from pre-response design variables and pre-treatment participant covariates. Post-response variables (RT, confidence) were explicitly excluded to prevent data leakage. Two participants who failed the vigilance check are excluded throughout. All trials from the five repeated movies are excluded (is_repeat = 0 only); because no repeated trials remain, `is_repeat` is not a predictor in any model.

- **Included (pre-response):** condition, target_type, movie_duration, stimulus_id
- **Optional controls (pre-treatment):** age, gender, handedness, vision
- **Excluded (leakage / post-response):** resp.rt, conf_radio.response
- **Excluded (zero-variance after filtering):** is_repeat

Three complementary models are used:

1. **Model 3** — participant-level binomial GLM
2. **Model 1** — trial-level GEE logistic regression (8 nested models)
3. **GLMM** — trial-level binomial mixed model with random participant intercept (Variational Bayes)

11.2 Model 3: Participant-Level Binomial GLM

A participant-level binomial GLM (link: logit) was fitted with condition, age, and gender as predictors, using (successes, failures) counts as the response.

Term	**Coef**	p	**OR**
const	2.608	$< .001$	13.570
age	-0.035	.0610	0.966
condition_NB	0.262	$< .001$	1.299
gender_Male	-0.241	.0054	0.786

Table 10: Model 3 — Participant-level binomial GLM: coefficients and odds ratios (repeated movies excluded).

Key finding: NB condition significantly increases accuracy odds. Male gender is associated with lower accuracy odds. Age shows a borderline negative effect.

11.3 Model 1: Trial-Level GEE Logistic Regression

Eight candidate GEE models (Binomial family, exchangeable working correlation, robust SE) were fitted with trials clustered by participant. Models were compared by QIC (lower is better); unstable models (non-finite parameters / SEs) were flagged and excluded from best-model selection.

Model	QIC	Description
M1_stim_interaction	4889.55	Interaction + demographics + stimulus duration.
M1_stim_main	4889.90	Main effects + demographics + stimulus duration.
M1_demo_interaction	4909.07	Interaction + demographics.
M1_demo_main	4909.37	Main effects + demographics (age, gender, handedness, vision).
M1_core_interaction	4930.40	Condition × target_type interaction. The interaction term tests whether the BB-vs-EM accuracy difference is moderated by condition (NB vs AB).
M1_core_main	4930.71	Main effects: condition + target_type.
M1_movieFE_interaction *(unstable)*	—	Interaction + movie fixed effects.
M1_movieFE_main *(unstable)*	—	Main effects + demographics + duration + movie fixed effects.

Table 11: Model 1 — GEE model comparison by QIC (lower is better; repeated movies excluded).

11.4 Best Model: M1_stim_interaction — Odds Ratios

The best stable model by QIC is M1_stim_interaction.

Term	Coef	SE	z	p	OR	95% CI
Intercept	1.916	0.588	3.260	.0011	6.796	[2.148, 21.505]
Condition: AB vs NB	-0.333	0.119	-2.791	.0053	0.717	[0.567, 0.906]
Target type: EM vs BB	0.256	0.101	2.535	.0113	1.292	[1.060, 1.576]
Gender: Male vs Female	-0.236	0.118	-2.004	.0451	0.790	[0.627, 0.995]
Handedness: Right	0.269	0.249	1.078	.2811	1.309	[0.802, 2.134]
Vision: Normal	0.043	0.096	0.448	.6542	1.044	[0.864, 1.261]
Vision: Uncorrected difficulty	1.171	0.396	2.958	.0031	3.224	[1.484, 7.002]
Condition(AB) × Target(EM)	0.115	0.141	0.817	.4139	1.122	[0.851, 1.479]
Age	-0.038	0.023	-1.631	.1028	0.963	[0.920, 1.008]
Movie duration (s)	0.018	0.003	5.129	$< .001$	1.018	[1.011, 1.025]

Table 12: Best GEE model: odds ratios with 95% CI (repeated movies excluded).

11.5 Interaction Effects in the GEE Models

Several candidate models include a **condition × target_type interaction**. This section explains how to interpret these interaction terms.

11.5.1 What the interaction term tests

The term `C(condition)[T.AB]:C(target_type)[T.EM]` tests whether the difference in log-odds of a correct response between EM-frame and BB-frame targets is *the same* in both the AB and NB conditions.

Let β_1 = coefficient for condition (AB vs NB, reference = NB), β_2 = coefficient for target type (EM vs BB, reference = BB), and β_3 = interaction coefficient. Then:

$$\begin{aligned}
\text{NB} + \text{BB}: &\quad \alpha \quad \text{(reference cell)} \\
\text{NB} + \text{EM}: &\quad \alpha + \beta_2 \quad \text{(EM advantage in NB)} \\
\text{AB} + \text{BB}: &\quad \alpha + \beta_1 \quad \text{(AB penalty for BB targets)} \\
\text{AB} + \text{EM}: &\quad \alpha + \beta_1 + \beta_2 + \beta_3 \quad \text{(combined effects in AB)}
\end{aligned}$$

So β_3 measures **how the EM advantage over BB changes** when moving from NB to AB.

11.5.2 Interpreting the sign of the interaction

- $\beta_3 > 0$ (OR > 1): the EM advantage is *larger* in AB than in NB — the AB condition hurts BB recall proportionally more than EM recall.
- $\beta_3 < 0$ (OR < 1): the EM advantage is *smaller* in AB — boundary disruption reduces BB and EM recognition approximately equally.
- $\beta_3 \approx 0$ (OR ≈ 1): no moderation — condition and frame type act independently.

11.5.3 Odds ratios for interaction terms are "ratios of odds ratios"

The OR for the interaction is

$$\text{OR}_{\text{interaction}} = e^{\beta_3} = \frac{\text{OR}_{\text{EM/BB in AB}}}{\text{OR}_{\text{EM/BB in NB}}}.$$

This is not a standalone probability but the *multiplicative change* in the EM advantage when switching from NB to AB.

11.5.4 Recovering simple (conditional) effects

- Effect of **condition for BB targets:** e^{β_1} (AB vs NB, BB trials only)
- Effect of **condition for EM targets:** $e^{\beta_1+\beta_3}$ (AB vs NB, EM trials only)
- Effect of **target type in NB:** e^{β_2} (EM vs BB, NB participants)
- Effect of **target type in AB:** $e^{\beta_2+\beta_3}$ (EM vs BB, AB participants)

These simple effects answer: "What is the effect of X when Y is held at a specific level?"

11.5.5 GEE vs GLMM for interaction estimates

GEE estimates *population-average* ORs ("On average across all participants, how does condition moderate the frame-type effect?"), while GLMM estimates *subject-specific* ORs ("For a given participant, how does condition moderate the effect?"). Both families should agree in sign and significance. A large discrepancy signals substantial between-participant heterogeneity in the moderation.

11.6 Diagnostics

11.6.1 Collinearity (VIF)

Variable	VIF
const	165.120
gender_Male	1.092
age	1.090
vision_Normal	1.063
condition_NB	1.050
vision_Uncorrected vision difficulty	1.025
handedness_Right handed	1.017
movie_duration	1.002
target_type_EM	1.002

Table 13: Variance inflation factors for trial-level predictors (VIF $< 5 =$ acceptable).

11.6.2 Heteroscedasticity (Breusch–Pagan, OLS surrogate)

A Breusch–Pagan test was run on an OLS surrogate model predicting participant-level accuracy from condition, age, and gender. The primary Model 1 is logistic (GEE) and does not assume homoscedasticity; this diagnostic is provided for completeness.

Model	BP LM	$p_{\mathbf{LM}}$	BP F	p_F	n	R^2
OLS accuracy	2.633	.4517	0.870	.4577	168	0.083

Table 14: Breusch–Pagan heteroscedasticity test (OLS surrogate on participant accuracy).

Both Breusch–Pagan statistics are non-significant ($p > .40$), indicating no evidence of heteroscedasticity in the OLS surrogate.

11.7 Mixed-Effects Logistic Regression (GLMM)

11.7.1 Rationale and Model Specification

Because each participant completed 35 trials (after repeated-movie exclusion), observations are not independent. A binomial mixed-effects logistic model addresses this by including a **random intercept for participant_id**, which controls for stable individual differences in baseline recognition accuracy.

Estimation uses **Variational Bayes** (`statsmodels BinomialBayesMixedGLM.fit_vb()`). Variational Bayes is preferred over the Laplace approximation (`fit_map`) because it reliably estimates the random-effect variance in balanced designs; the Laplace approximation can collapse the random-effect SD to zero in such cases. Approximate z-statistics and p-values are derived from $z = \hat{\beta}/\text{Post.SD}$ using the standard normal distribution.

Three GLMM variants are reported:

1. **Core** (condition + target_type main effects)
2. **Interaction** (condition × target_type)
3. **Full** (condition + target_type + age + gender + movie_duration)

11.7.2 Fixed Effects

Predictor	Post. Mean	Post. SD	z	p (approx)	OR
Intercept	1.8022	0.0370	48.717	$< .001$	6.0632
Condition: AB vs NB	-0.2816	0.0511	-5.509	$< .001$	0.7546
Target type: EM vs BB	0.3129	0.0562	5.571	$< .001$	1.3674

Table 15: GLMM core — main effects (repeated movies excluded).

Predictor	Post. Mean	Post. SD	z	p (approx)	OR
Intercept	1.8270	0.0370	49.377	$< .001$	6.2150
Condition: AB vs NB	-0.3288	0.0512	-6.425	$< .001$	0.7198
Target type: EM vs BB	0.2560	0.0562	4.559	$< .001$	1.2918
Condition(AB) $\times$ Target(EM)	0.1112	0.0782	1.422	.1549	1.1176

Table 16: GLMM interaction — condition $\times$ target_type (repeated movies excluded).

Predictor	Post. Mean	Post. SD	z	p (approx)	OR
Intercept	2.0576	0.0371	55.497	$< .001$	7.8273
Condition: AB vs NB	-0.2925	0.0513	-5.706	$< .001$	0.7464
Target type: EM vs BB	0.3269	0.0563	5.809	$< .001$	1.3867
Gender: Male vs Female	-0.2551	0.0422	-6.047	$< .001$	0.7749
Age	-0.0305	0.0017	-18.385	$< .001$	0.9699
Movie duration (s)	0.0186	0.0011	16.819	$< .001$	1.0187

Table 17: GLMM full — main effects + age + gender + movie_duration (repeated movies excluded).

11.7.3 Random Effects

The random-effect standard deviation (SD) quantifies between-participant variability in baseline log-odds accuracy, after accounting for fixed effects.

GLMM variant	Random intercept SD	Interpretation
Core (main effects)	0.4203	Between-participant variability in log-odds
Interaction	0.4205	Same, after adding interaction
Full (+ age, gender, duration)	0.4011	Same, after demographic controls

Table 18: GLMM random intercept SDs (log-odds scale; repeated movies excluded).

A random intercept SD of ≈ 0.42 indicates meaningful between-person variability in baseline recognition accuracy — justifying the inclusion of random intercepts per participant. The SD is stable across model variants, suggesting it is not confounded with the fixed-effect predictors.

11.7.4 Interaction Interpretation in the GLMM

The interaction term in the GLMM (`C(condition)[T.AB]:C(target_type)[T.EM]`) has the same structural interpretation as in the GEE (see §11.5), but now represents a *subject-specific* (conditional) effect rather than a population-average effect.

The posterior mean of the interaction coefficient indicates whether the EM advantage over BB is larger or smaller in the AB condition relative to NB, *for a given participant* (after integrating out individual differences via the random intercept).

Key comparison with GEE:

- If GEE and GLMM interaction ORs agree in sign and magnitude: the moderation is consistent across participants.
- If they diverge substantially: there is heterogeneity in how the condition moderates the frame-type effect across individuals.

11.7.5 Summary

The GLMM provides the most methodologically rigorous answer to the primary research question. After controlling for individual participant ability, the Abrupt Boundary (AB) condition significantly reduces recognition accuracy relative to Natural Boundary (NB). This converges with the GEE and the participant-level H1 result, establishing consistency across all three levels of analysis.